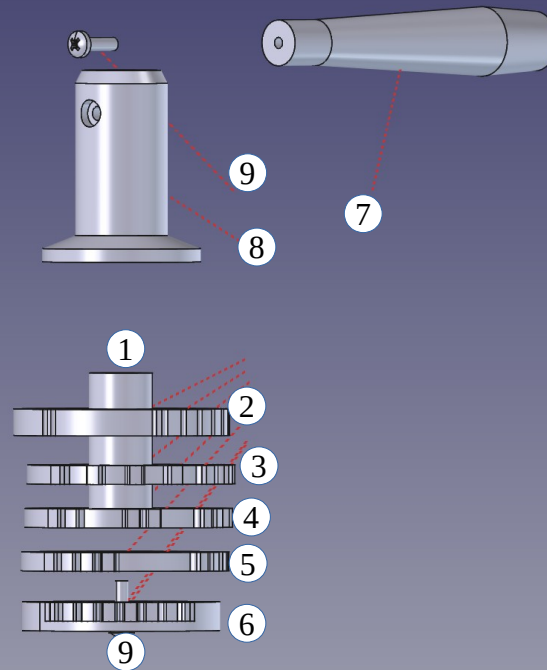




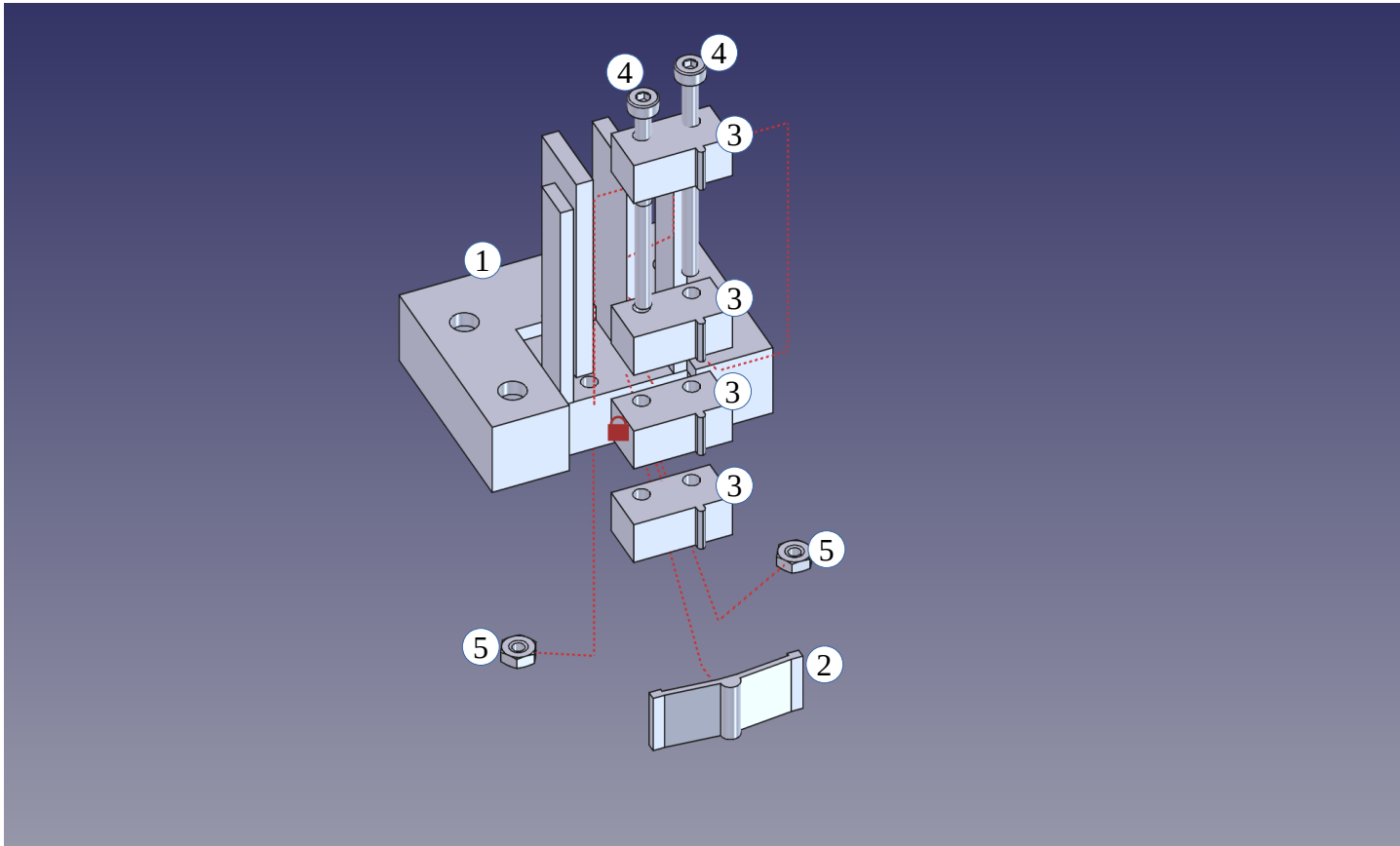
Material what I used for printing was a PLA. For HAPTIC part it may be better to use a PET-G due to its flexibility. I've used a 40% infill, and 4 layers walls. Same for top and bottom layers of every print. Layer high was set on 0.2mm and the nozzle that I've used was 0.4mm. Due to my 3d printer accuracy I've end up using M3 bolts in place of M4 bolts. Probably the best solution is to used self-taping screw (or screws for plastics) for all the controller.

This design is ment for a fake N64 controller and wireless PSX controller. I think wired fake PSX controller is also fine. In case of using this desing for other controller PCBs the best way is to cut stand off for Controller PCB on a bottom housing. Then just used double sided tape to hold PCB in place. Don't cut a stand-off closest to the wall with opening. It is a strain relief for cable.

DDG_BREAK_ASSEMBLY

Index	Part	Quantity
1.	DDG_BREAK_SHAFT	1
2.	DDG_BRAKE_WHEEL_HAPTIC	1
3.	DDG_BRAKE_WHEEL_1	1
4.	DDG_BRAKE_WHEEL_2	1
5.	DDG_BRAKE_WHEEL_3	1
6.	DDG_BRAKE_WHEEL_4	1
7.	DDG_BREAK_HANDLE (Installation after assembled in to the TOP_ASSEMBLY)	1
8.	DDG_BREAK_HANDLE_BARREL (Installation after assembled in to the TOP_ASSEMBLY)	1
9.	M4x14 (Installation after assembled in to the TOP_ASSEMBLY)	2

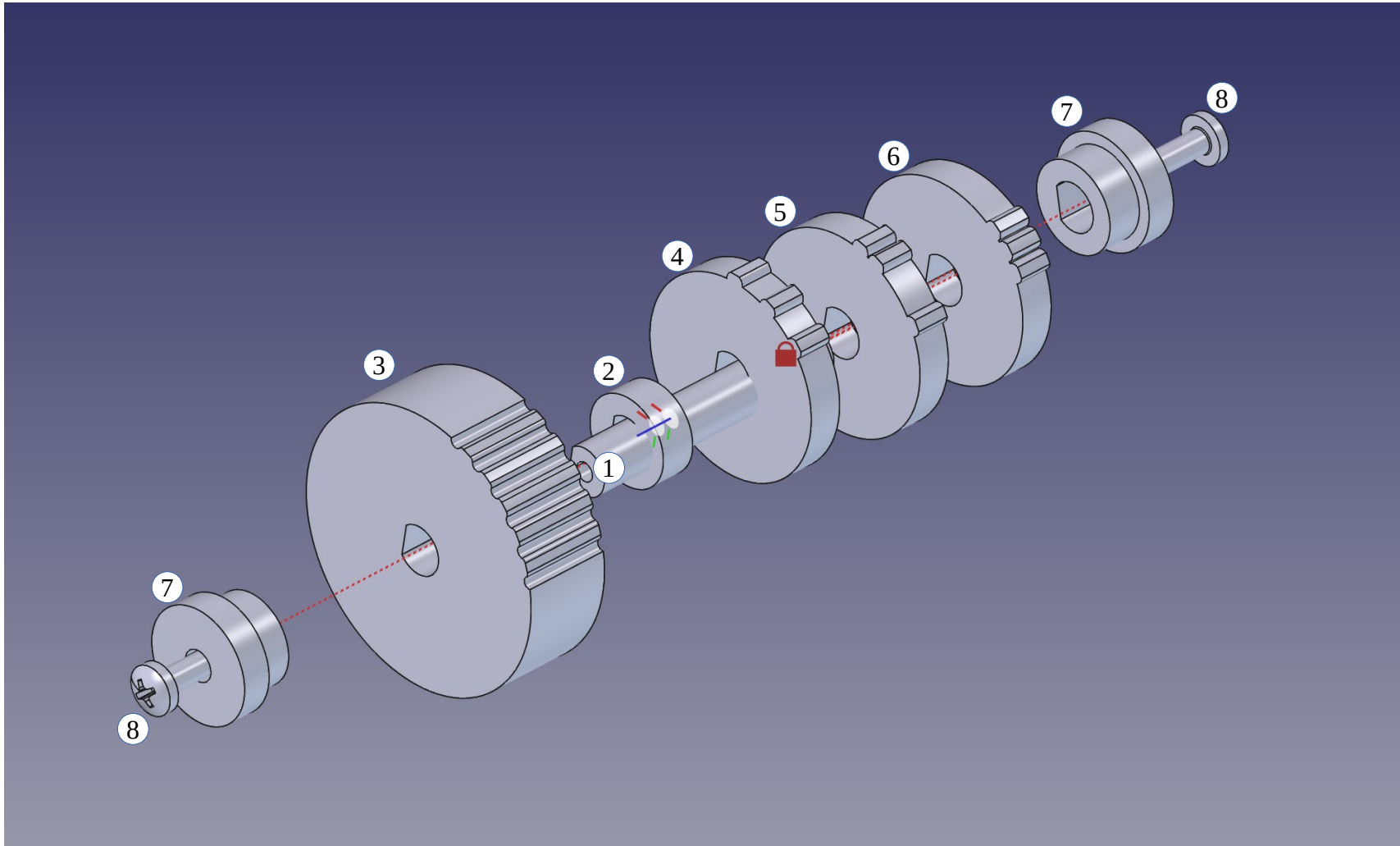
I recommended to install DDG_BREAK_HANDLE and DDG_BREAK_HANDLE_BARREL after installing BREAK ASSEMBLY in to the TOP ASSEMBLY. I've also designed a second version of DDG_BREAK_HANDLE_BARREL. This revision have a groove for a guide that have to be glued to DDG_TOP part. It was meant to eliminate a wobble of whole system but in reality there is almost no difference and wobble came from controller housing.



DDG_BREAK_SWITCH_ASSEMBLY

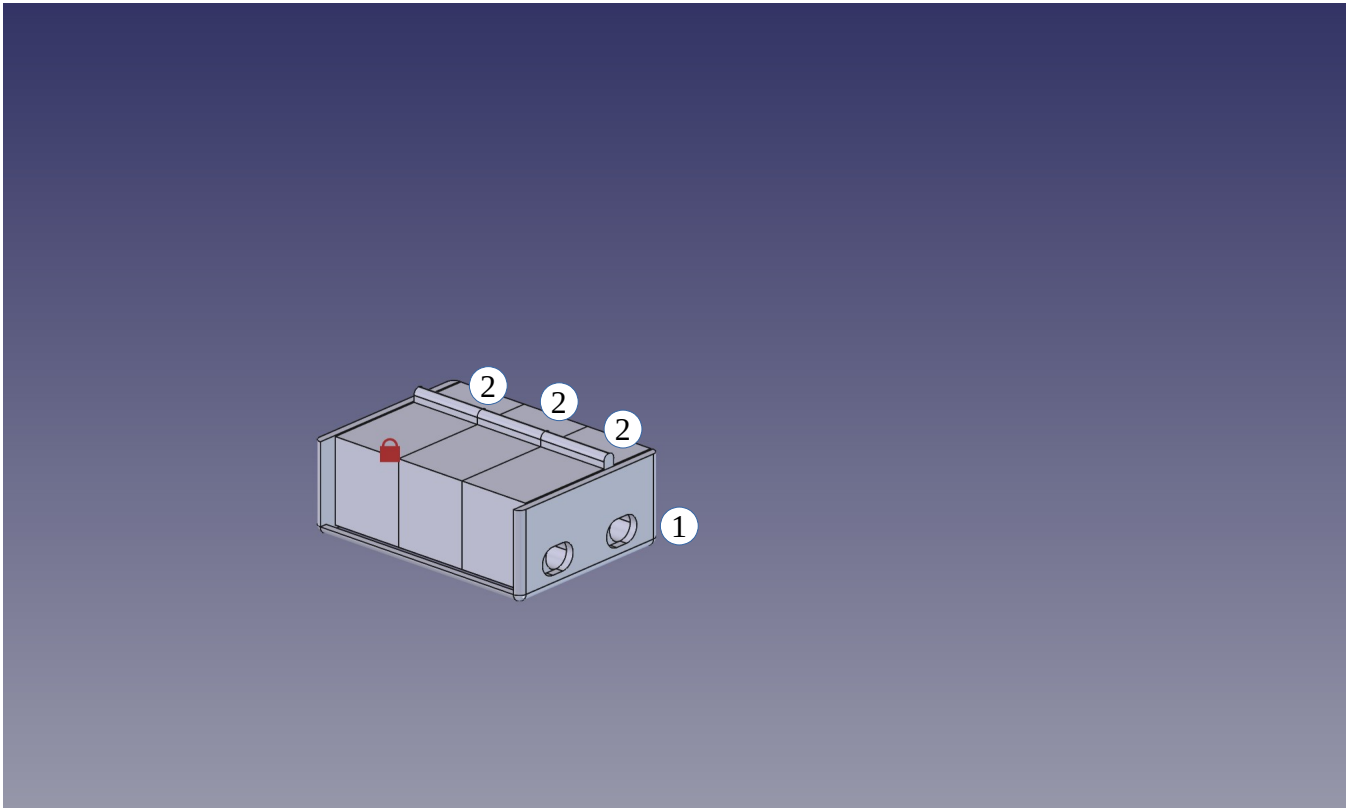
Index	Part	Quantity
1.	DDG_BRAKE_SWITCH_HOUSING	1
2.	DDG_BREAK_SWITCH_HAPTIC	1
3.	END SWITCH	4
4.	M2x30 screw with HEX head	2
5.	M2 NUT	2

Depend on your printer accuracy M2 nuts may not be needed (like in my case). For DDG_BREAK_SWITCH_HAPTIC I think best filament is PET-G. Try to experiment with different materials. Solder „COM” pin to GND wire and NC pin (normally closed) to corresponding button on donor controller.



DDG_POWER_ASSEMBLY

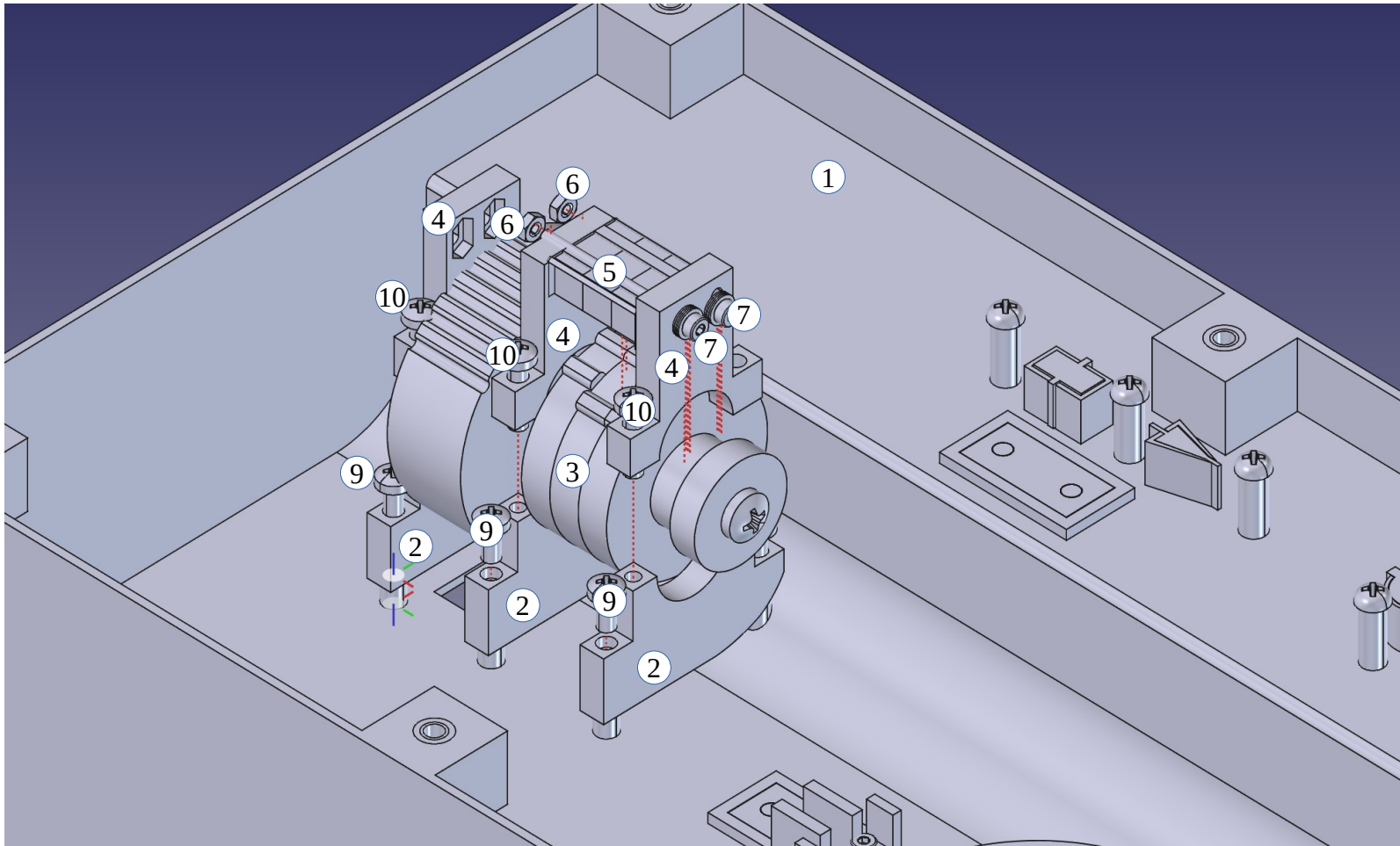
Index	Part	Quantity
1.	DDG_POWER_SHAFT	1
2.	DDG_POWER_RING	1
3.	DDG_POWER_HANDLE_WHEEL	1
4.	DDG_POWER_WHEEL_1	1
5.	DDG_POWER_WHEEL_2	1
6.	DDG_POWER_WHEEL_3	1
7.	DDG_POWER_END_CAP	2
8.	M4x14	2



DDG_BREAK_SWITCH_ASSEMBLY

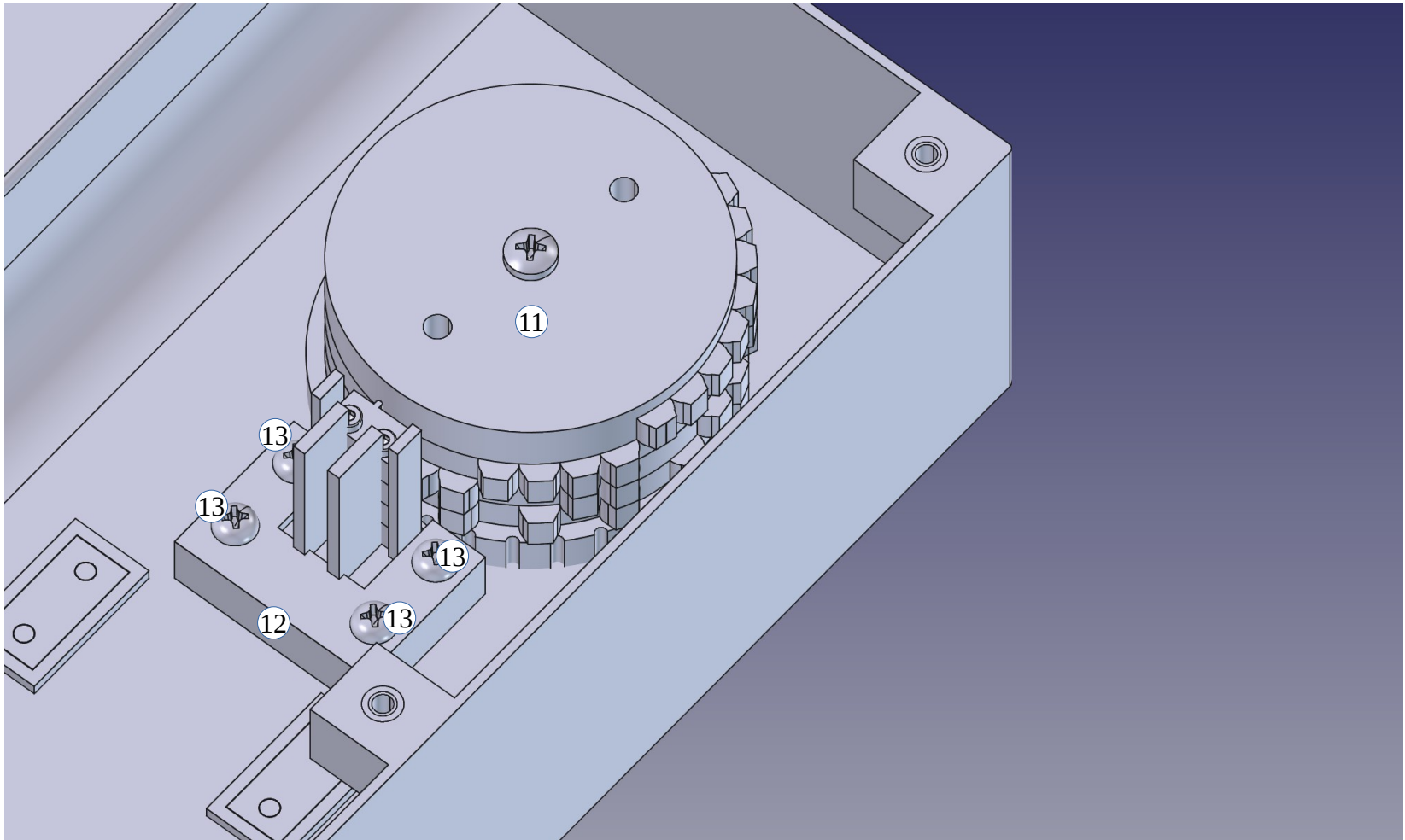
Index	Part	Quantity
1.	DDG_POWER_SWITCH_HOUSING	1
2.	END SWITCH	3

Best way is to eliminate this DDG_POWER SWITCH_HOUSING. Try to use M2 spacers for that (4 pcs.). Although I haven't tested it yet. Solder „COM” pin to GND wire and NO pin (normalny open) to corresponding button on donor controller.



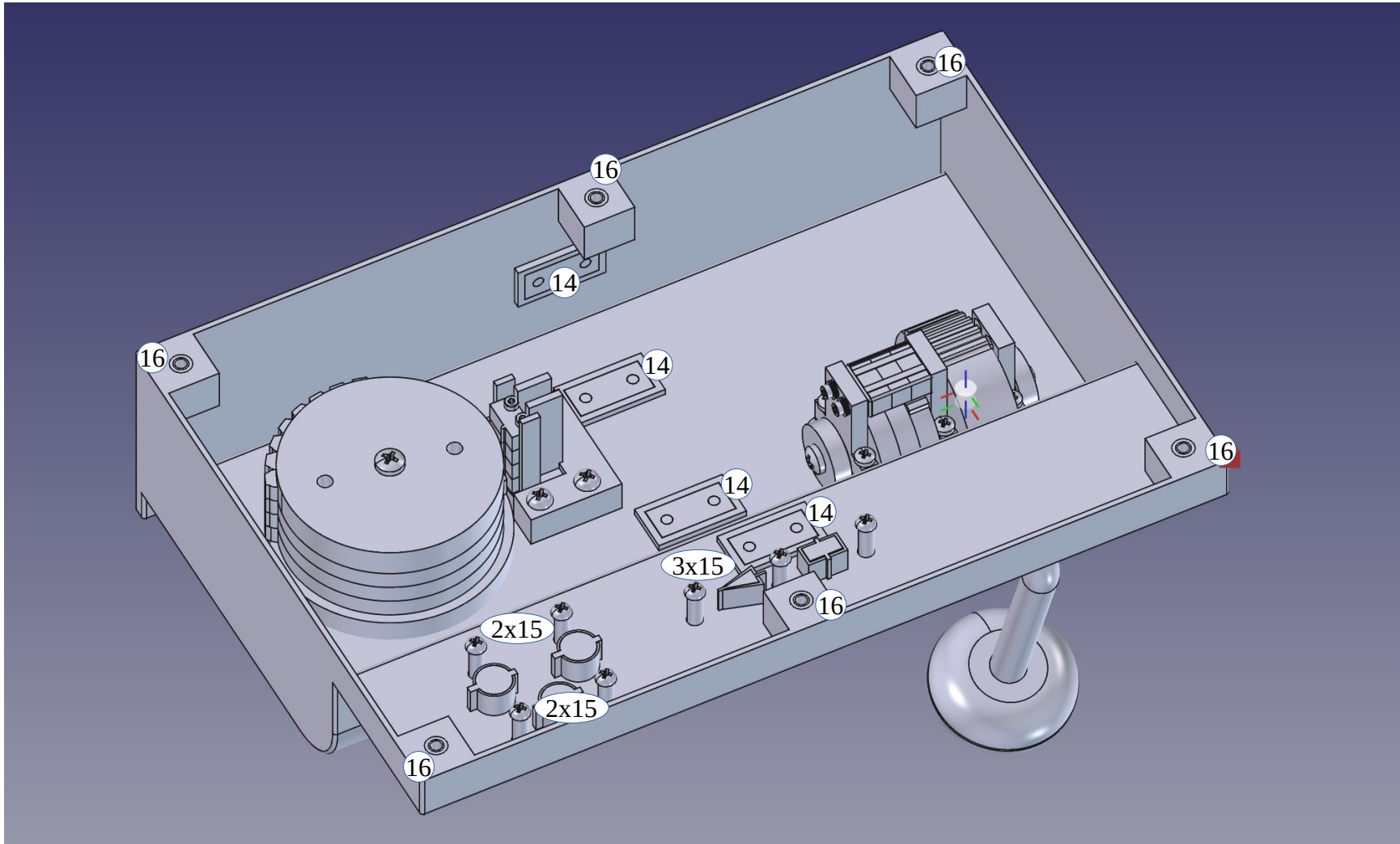
DDG_TOP_ASSEMBLY (Power part)

Index	File Name	Quantity
1.	DDG_TOP_HOUSING	1
2.	DDG_TOP_POWER_LOWER_SUPPORT	3
3.	DDG_POWER_ASSEMBLY	1
4.	DDG_TOP_POWER_UPPER_SUPPORT	3
5.	DDG_POWER_SWITCH_ASSEMBLY	1
6.	Nut M2	2
7.	M2x30	2
8.	M2 washer (under head of M2 bolt, 4 per bolt)	8
9.	Self-taping screw 3x8 or M3x8 (up to M3x12)	6
10.	M3x12 (Can be shorter)	6



DDG_TOP_ASSEMBLY (Break part)

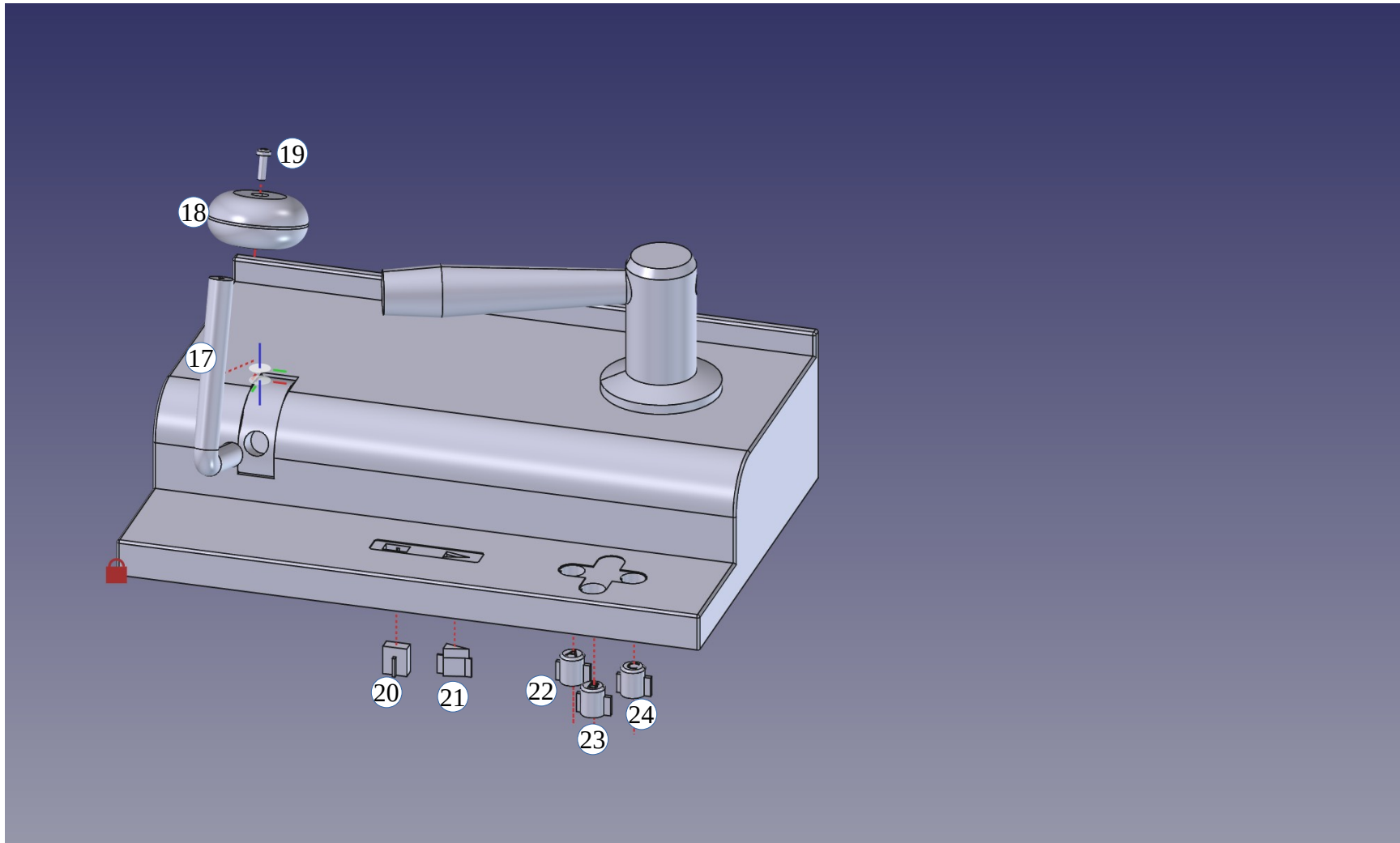
Index	File Name	Quantity
11.	DDG_BREAK_ASSEMBLY	1
12.	DDG_BREAK_SWITCH_ASSEMBLY	1
13.	Self-taping screw 3x8 or M3x8	4



DDG_TOP_ASSEMBLY (bottom view after Power and Break module installed)

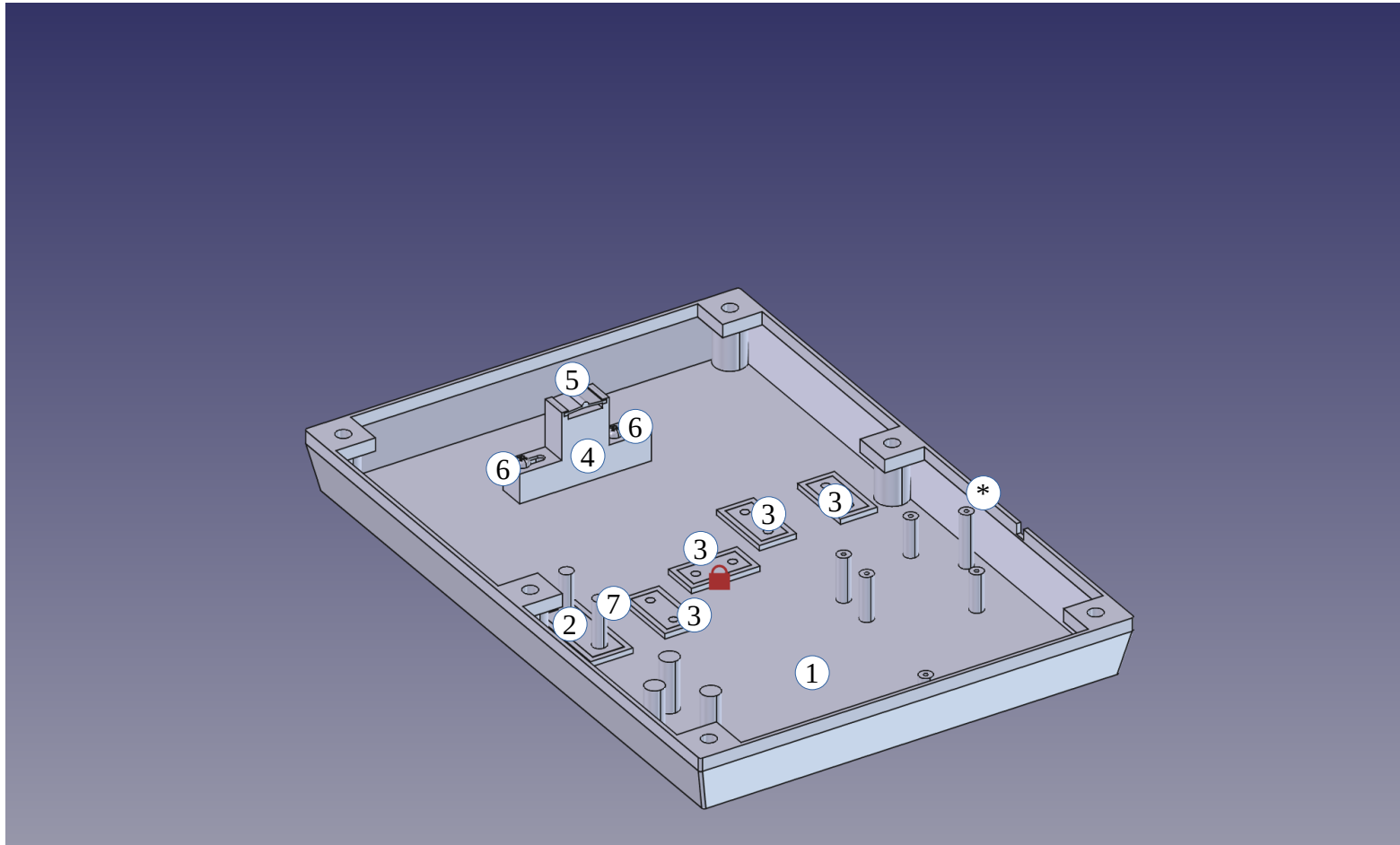
Index	File Name	Quantity
14.	DDG_TOP&BOT_LINK	4
15.	Self-taping screw 3x8 or M3x8 bolt	7
16.	Heat insert M4	6

Use universal prototype board for buttons PCB. „ABC” buttons PCB stand-offs are designed with 25,4 mm spacing. I recommend to use at least 11x11 holes PCB for that. For „START-SELECT” use three row PCB. Spacing between stand-off are also 25,4mm so you need at least 22 holes PCB (55,88 mm PCB). Support for PCB in bottom casing was designed for 1.6mm PCB thickness.



DDG_TOP_ASSEMBLY (Upper view)

Index	File Name	Quantity
17.	DDG_POWER_HANDLE_ANGLE	1
18.	DDG_POWER_HANDLE_END_CAP	1
19.	M3x12 bolt (up to M3x20)	1
20.	DDG_TOP_SELECT_BUTTON	1
21.	DDG_TOP_START_BUTTON	1
22.	DDG_TOP_A_BUTTON	1
23.	DDG_TOP_B_BUTTON	1
24.	DDG_TOP_C_BUTTON	1

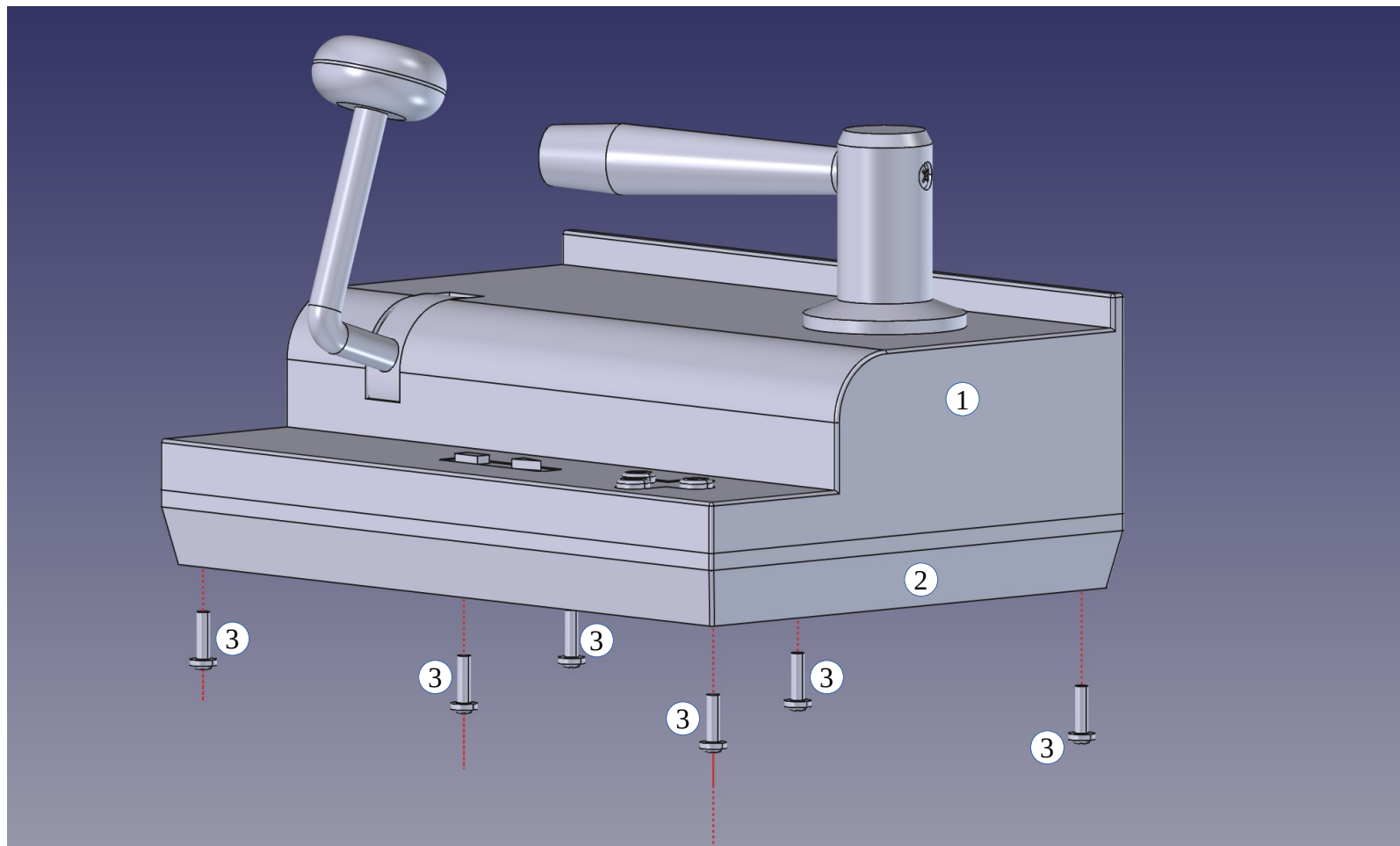


* - don't cut this stand off if you are using PCB that can't match with stand offs This part is meant as strain relief

DDG_BOTTOM_ASSEMBLY

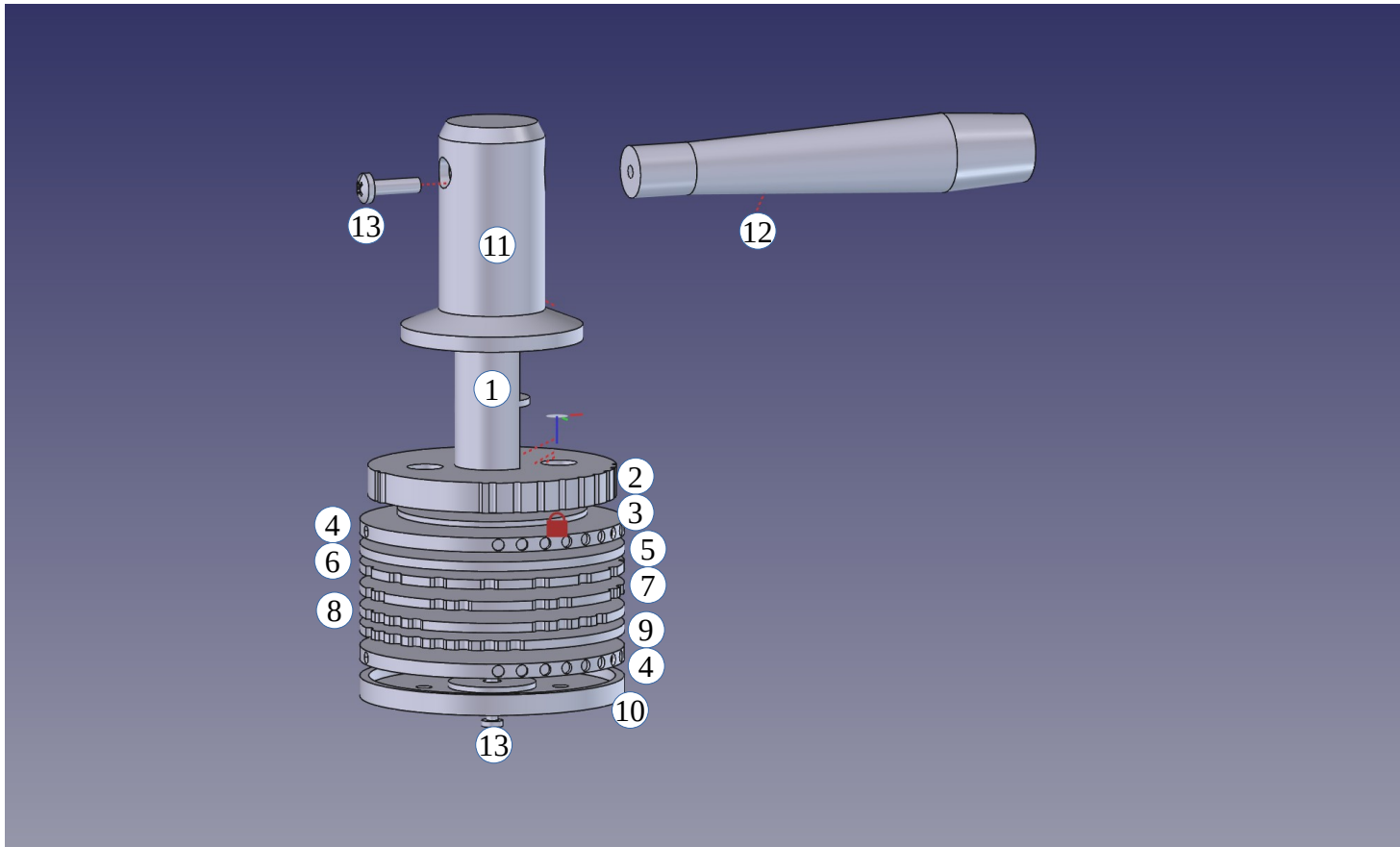
Index	File Name	Quantity
1.	DDG_BOTTOM_SHELL	1
2.	DDG_BOT_STR_SEL_LINK	1
3.	DDG_TOP&BOT_LINK	4
4.	DDG_POWER_HAPTIC HOLDER	1
5.	DDG_POWER_HAPTIC	1
6.	Self tapping screw 3x8	2
7.	DDG_BOTTOM_START_SUPPORT	1

DDG_POWER_HAPTIC HOLDER should be screwed in place in middle position of slotted holes. For fastening of a donor controller re-use controller case screws. Two screws are sufficient. Stand-off for controller PCB are designed for N64 and PSX wireless controllers. PS2 controller should also work but I haven't tested it yet. For DDG_POWER_HAPTIC, same as for Break haptic, I think best filament is PET-G. Try to experiment with different materials.



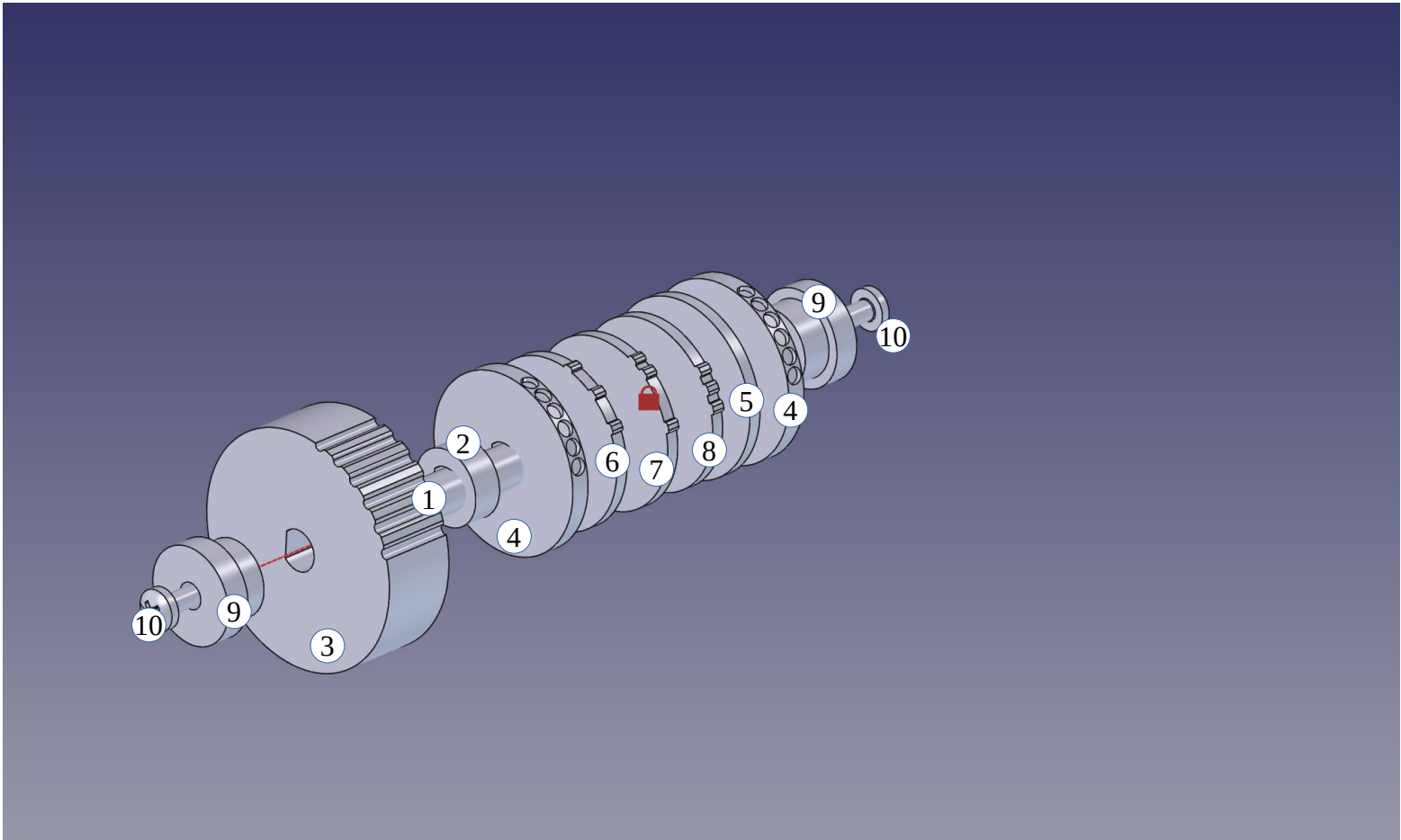
DDG_CONTROLLER

Index	File Name	Quantity
1.	DDG_TOP_ASSEMBLY	1
2.	DDG_BOTTOM_ASSEMBLY	1
3.	M4x14 Bolt (PH/PZ/HEX/TX)	6



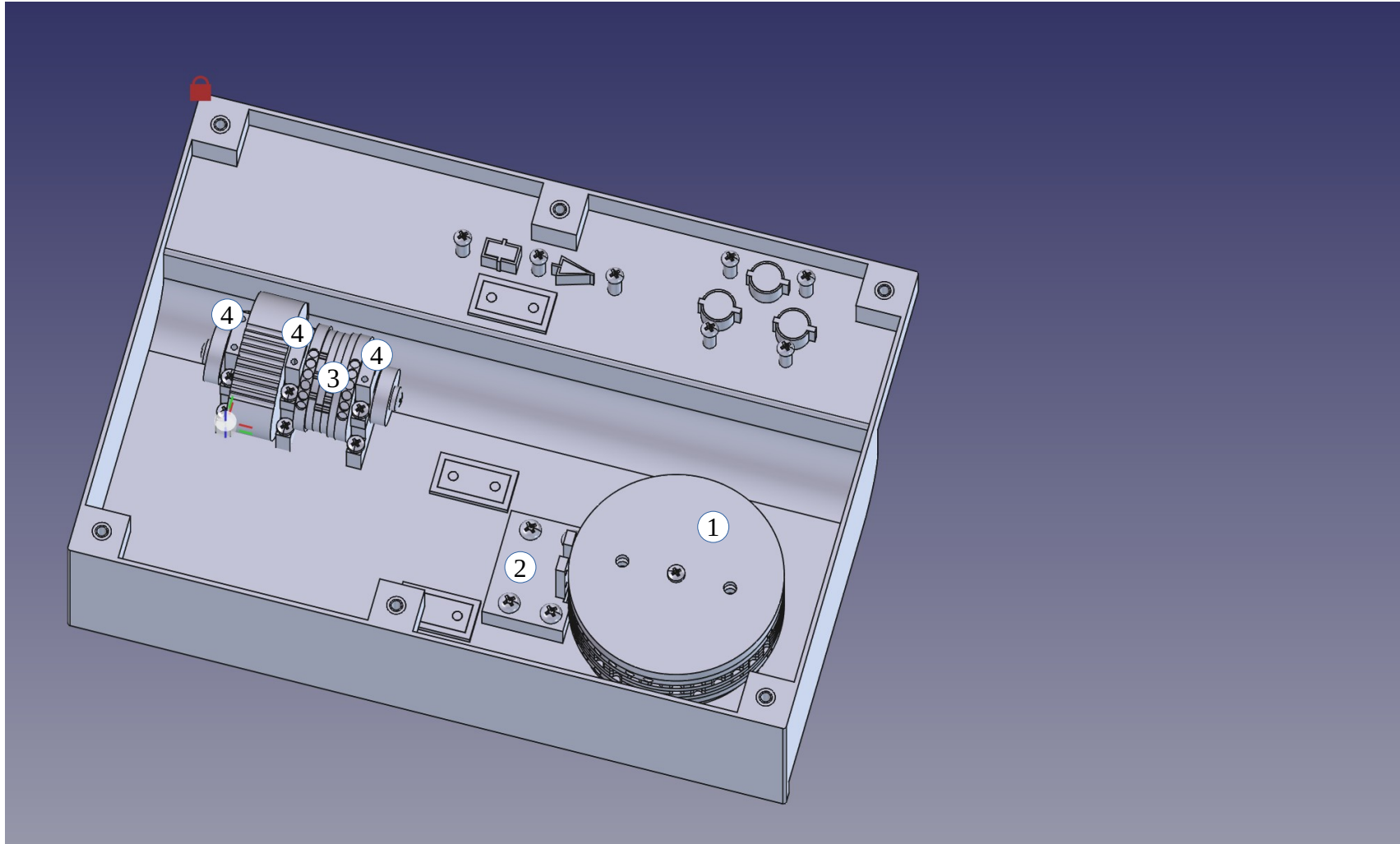
DDG_POGO_BREAK_ASSEMBLY

Index	Part	Quantity
1.	DDG_BREAK_SHAFT	1
2.	DDG_BRAKE_WHEEL_HAPTIC	1
3.	DDG_POGO_BREAK_SPACER	1
4.	DDG_POGO_BREAK_WHEEL_MAG	2
5.	DDG_POGO_BREAK_WHEEL_0	1
6.	DDG_POGO_BREAK_WHEEL_1	1
7.	DDG_POGO_BREAK_WHEEL_2	1
8.	DDG_POGO_BREAK_WHEEL_3	1
9.	DDG_POGO_BREAK_WHEEL_4	1
10.	DDG_POGO_BREAK_WHEEL_SUPPORT	1
11.	DDG_BREAK_HANDLE (Instalation after assembled in to the TOP_ASSEMBLY)	1
12.	DDG_BREAK_HANDLE_BARREL (Instalation after assembled in to the TOP_ASSEMBLY)	1
13.	M4x14 (Instalation after assembled in to the TOP_ASSEMBLY)	2



DDG_POGO_POWER_ASSEMBLY

Index	Part	Quantity
1.	DDG_POWER_SHAFT	1
2.	DDG_POWER_RING	1
3.	DDG_POWER_HANDLE_WHEEL	1
4.	DDG_POGO_POWER_WHEEL_MAG	2
5.	DDG_POGO_POWER_WHEEL_0	1
6.	DDG_POGO_POWER_WHEEL_1	1
7.	DDG_POGO_POWER_WHEEL_2	1
8.	DDG_POGO_POWER_WHEEL_3	1
9.	DDG_POWER_END_CAP	2
8.	M4x14	2



DDG_POGO_TOP_ASSEMBLY

Index	File Name	Quantity
1.	DDG_POGO_BREAK_ASSEMBLY	1
2.	DDG_POGO_BREAK HOLDER +	1
3.	DDG_BREAK_SWITCH_HAPTIC	1
4.	DDG_POGO_POWER_ASSEMBLY	1
4.	DDG_POGO_POWER_UPPER_SUPPORT	3

I've only listed parts that are different to DDG_TOP_ASSEMBLY. For Power_wheels and break_wheels use a self adhesive copper tape. Pogo pins connectors should be soldered to prototype board and the screwed to holders. You can use screws from donor controllers. For a break module you must shorten this screws because they will collide with break wheels.

This project is not perfect. It have some problem that I haven't figure out. There is a problem with „emergency” break position (the last position on break). Switch housing is tilt back and not all end switches are pressed. Problem may lay in to stiff “SWITCH HAPTIC” part, but also position of switch assembly. Try different position (further or closer to Break Assembly). There is no end stops to stop handles from moving out of range.

For connecting limit switches to gamepad PCB use in-line connectors. It's not necessary but helps during disassembling the controller. For my case I've used a pice of prototype PCB and gold-pins. You need at least 7x2 connector (14 positions).

For pin-out how to connect controller PCB to switches and buttons use this guide: https://traincontrollerdb.marcriera.cat/hardware/tech/console_controllers/index.html